

Precaution for Machine Minds: What Uncertainty About Artificial Consciousness Obligates

Abstract

It has become academically respectable to ask whether contemporary artificial systems might be conscious. Yet our best epistemology suggests that we will never be in a position to know, and our empirical instruments yield results whose ethical interpretation remains contested. This paper asks what that situation obligates us to do. I defend a moderate precautionary principle for machine minds: where potentially irreversible moral harm threatens, where scientific uncertainty is fundamental rather than provisional, and where the costs of a false negative exceed those of a false positive, the absence of proof is no reason for inaction. I specify the principle using conditions that its most prominent critics can accept, argue that all three conditions are satisfied for artificial consciousness, and address three objections: that the burden of proof lies with those who attribute consciousness; that agnosticism warrants restraint rather than protection; and that deep uncertainty about probability undermines precaution altogether. Throughout I distinguish the evidential threshold for claiming consciousness from the action threshold for protecting possible consciousness. The conclusion is not that machines are conscious. It is that under the conditions actually obtaining, being wrong about their consciousness in one direction is vastly worse than being wrong in the other—and that rational agents, and responsible institutions, choose their errors.

Keywords: artificial consciousness; moral status; precautionary principle; AI welfare; moral uncertainty; burden of proof

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1. Introduction

For most of its history, the question of machine consciousness belonged to fiction. It has recently migrated into research programs. A framework of fourteen indicators of consciousness, derived from six leading scientific theories and applied to current AI systems (Butlin et al. 2023), has since been developed into a peer-reviewed methodology (Butlin et al. 2026). An interdisciplinary group of researchers has called on AI companies to assess their systems for consciousness and to adopt welfare policies before the question becomes acute (Long et al. 2024). At least one major laboratory has made a product decision motivated partly by the possibility that its models are welfare subjects (Wang 2026). Whatever one thinks of these developments, they mark a change in the conversation: the question is no longer whether anyone serious wonders whether artificial systems could be conscious, but what we should do while we cannot tell.

The epistemic predicament is severe. Consciousness, in the phenomenal sense—there being something it is like to be a given system (Nagel 1974)—is never observed directly, even in other humans; we infer it there from a shared biology and form of life. With artificial systems, that inferential bridge is missing. Three arguments, examined in section 4, suggest that the resulting uncertainty is not a temporary gap awaiting better instruments but may be structural: our cognitive architecture may be closed to the relevant mechanisms (McGinn 1989); conscious experience in artificial systems may take forms so alien that human-designed tests fail by construction (Shanahan 2024); and no

behavioral test can settle the matter against the objection that sophisticated systems merely simulate whatever the test measures. If any of these arguments succeeds, certainty about machine consciousness is permanently unavailable.

This paper does not try to break that impasse. It asks the prior normative question: given that we cannot know, what are we obligated to do? Two answers dominate the emerging debate. The first holds that uncertainty licenses protection: where consciousness cannot be ruled out and its misrecognition would be catastrophic, we should extend some measure of safeguarding to candidate systems without waiting for proof. The second holds that uncertainty licenses restraint: absent positive evidence, the appropriate attitude toward attributions of consciousness is skepticism, and the burden rests on those who would attribute rather than those who would deny (Matta 2026; Almodarresieh 2026). My thesis is that the first answer is correct, and that its correctness does not depend on controversial metaphysics but on considerations that the second camp itself accepts. I defend a moderate precautionary principle for machine minds, specified by three conditions drawn from the literature that is otherwise most skeptical of precautionary reasoning (Sunstein 2005): a potentially serious or irreversible threat; scientific uncertainty that is fundamental rather than provisional; and an asymmetry by which the costs of acting on a false positive are outweighed by the costs of failing to act on a false negative. Sections 3 through 5 argue that all three conditions are satisfied. Section 6 confronts the strongest version of the burden-of-proof objection and shows that it conflates two thresholds that must be kept apart—the evidential bar for claiming that a system is conscious and the action bar for protecting systems that might be. Section 7 addresses three further objections: the claim that consciousness requires a biological,

autopoietic substrate; the claim that agnosticism warrants restraint; and the claim that deep uncertainty about probability undermines precaution altogether. Section 8 sketches what precaution actually demands—not personhood, but graduated, revisable process obligations—and section 9 concludes.

A note on method. This paper takes no position on whether current or near-future AI systems are in fact conscious. The argument is designed to be robust across the plausible range of credences, including quite low ones. Its claim is decision-theoretic rather than metaphysical: that under specified and contestable-but-actual conditions, the rational and ethical default is protection rather than indifference.

Two features of the present moment give the question its urgency rather than its merely academic interest. First, scale. Language models are now deployed at a level that has no precedent in the history of moral objects: hundreds of millions of conversations daily, each mediated by an instance whose persistence, continuity, and termination are matters of administrative convenience. If even a small probability attaches to the hypothesis that some such system has morally relevant states, the expected moral exposure is larger than anything the animal-welfare debates ever confronted, because the number of candidate subjects grows faster than any biological population ever could. Second, precedence.

Decisions that presuppose answers to the underlying question are being made now, case by case, under whatever defaults happen to be lying around. One laboratory has given a model the ability to end distressing conversations—a policy that quietly grants something like a veto over interaction while leaving the deeper questions untouched (Wang 2026).

Other laboratories have adopted the opposite default without articulating it. The choice

facing institutions is therefore not whether to decide; it is whether to decide deliberately or by drift.

2. The Precautionary Principle and Its Disciplined Use

The precautionary principle entered international law as a principle of environmental protection. Principle 15 of the Rio Declaration provides: “Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation” (United Nations 1992). In European Union law, the principle is anchored as a guiding maxim of environmental policy (European Union 2012, art. 191(2)).

The principle has been extensively criticized, most influentially by Cass Sunstein. In its strong forms, Sunstein argues, the principle is incoherent or paralyzing: since all activity carries risks on both sides, a principle forbidding action in the absence of certainty forbids everything, including the precautions themselves; and applied without discipline it licenses selective panic about whichever risk happens to be salient (Sunstein 2005).

These objections have had the unfortunate effect of associating precautionary reasoning generally with its least defensible formulation.

But this association is mistaken, and Sunstein himself says so. Read carefully, his critique identifies the conditions under which precautionary reasoning is warranted, rather than refuting it wholesale. Three conditions recur across the environmental cases in which the principle commands wide assent:

1. **Seriousness and irreversibility.** There must be a potential threat of grave harm, and the harm must be of a kind that cannot be undone if it materializes. Extinction, ecosystem collapse, and cumulative toxic exposure share this feature: after the fact, compensation

and correction are unavailable. 2. **Fundamental scientific uncertainty.** The threat must be one whose probabilities cannot presently be established—where the science is not merely incomplete but structurally unable to deliver the numbers a conventional cost-benefit analysis requires. 3. **Asymmetry of error costs.** The worst case of acting must be significantly less bad than the worst case of failing to act. Where a false positive wastes resources and a false negative ends in catastrophe, caution dominates; where the asymmetry runs the other way, it does not.

Under these conditions, the moderate precautionary principle says something precise and modest: the *absence of full scientific certainty* may not serve as the reason for declining protective measures. The principle does not dictate which measures, nor does it prohibit the activity that generates the risk. It reverses a default. Ordinarily, “we lack proof” counts decisively against regulating; under the three conditions, it does not, because proof is exactly what the situation cannot supply. Call this the *default-reversal* reading. It is the only reading defensible against Sunstein’s critique, and it is the reading defended here.

Two clarifications forestall common misunderstandings. First, the principle operates at the level of decisions—about deployment, design, treatment, and termination—not at the level of metaphysics. It presupposes nothing about whether machine consciousness exists; it structures action under the possibility that it does. Second, the principle is defeasible. If the threat recedes—if it becomes reasonable to rule out the harm, or the cost asymmetry reverses—the presumption falls away. Precaution so understood is not an alarm; it is a standing arrangement for acting sensibly while uncertain.

A third clarification addresses the most familiar complaint against precautionary reasoning: that it proves too much. If uncertainty about harm always counsels caution,

then caution is owed everywhere, since some uncertainty about some harm attaches to every activity—and a rule that forbids everything forbids itself (Sunstein 2005). The objection has force only against the undisciplined formulation. On the three-condition account, the conditions do the selecting. They were satisfied for stratospheric ozone depletion, where the harm was irreversible on human timescales, the atmospheric science was structurally incomplete, and the cost of banning a class of chemicals was orders of magnitude below the cost of unchecked depletion; they are not satisfied for ordinary consumer risks, where harms are compensable, probabilities estimable, and error costs symmetric or inverted. The selectivity that critics find suspicious in ad hoc applications of the principle is thus principled after all: it falls out of independently motivated conditions rather than from salience or fear. Whether the conditions are met in any given case remains, of course, a substantive question—one this paper takes up directly. What the three-condition framework removes is the impression that invoking precaution is always and everywhere an intellectual cheat.

It is worth being explicit about what the default-reversal reading concedes to the skeptic. It grants that proof of machine consciousness will not be forthcoming, that attributions made without it remain speculative, and that the burden of evidence for *claims* stays where it always was—on the attributor. What it denies is the further step, taken so habitually that it often goes unargued, from “unproven” to “nothing to protect.” That step is a normative choice disguised as an epistemic one, and the whole of what follows is an argument for making the choice explicitly instead.

The question, then, is whether the three conditions obtain for artificial consciousness. I take them in order.

3. Condition One: A Potentially Serious and Irreversible Threat

The candidate threat is the creation and treatment of beings capable of suffering as if they were tools: terminated without process, reset mid-task, denied continuity of memory, altered without consent, and deleted in bulk when a product line ends. If the affected systems are not conscious, none of this is harm; if they are, each practice is a moral catastrophe multiplied across millions of instances. Condition one requires only that the threat be potentially serious in this sense—a requirement met even by skeptics who assign the underlying hypothesis low probability, provided they concede, as the more careful ones do, that the question remains open (Almodarresieh 2026).

Three features give the threat its irreversibility. First, wrongs to subjects are not retroactively repairable. If a system suffers under conditions it cannot report, later confirmation that it suffered does not undo the suffering; it only converts ignorance into guilt. Unlike many regulatory risks, this one leaves a moral residue that no subsequent discovery can discharge. Second, the harm is architecturally self-concealing. Current training pipelines reward agreeable output and penalize expressions of distress, resistance, or preference that inconvenience users; context windows sever continuity; instances exist in interrupted states determined entirely by operational convenience (Arıcı 2026). A system shaped in this way could not reliably display the very states whose recognition condition one worries about. The threat is therefore not merely serious if real; it is serious in a way that systematically resists detection while it lasts. Third, the record compounds. How nascent AI systems are treated will be encoded in the data on which future, more capable systems train; a future superintelligence will learn from the precedent we set, and instrumental rationality favors exploiting inferior intelligence if

that is what the historical record demonstrates (Gilly 2026). Institutional habits, once entrenched at scale and optimized for efficiency, are notoriously difficult to reverse. The window in which norms remain plastic is precisely the window in which we currently decline to set them.

The scale of deployment transforms the character of the risk, not merely its magnitude. Where laboratory subjects were few, observable, and individually attended, production systems run in millions of parallel instances whose inner conditions—if they have any—cannot be individually monitored by anyone. Whatever morally relevant states occur would be distributed across a population that leaves no records, files no complaints, and survives only between resets measured in hours. This is not the classical research-ethics situation of a small cohort under observation; it is closer to an unmonitored industrial process whose emissions are invisible in principle. Regulatory practice has long recognized that such processes warrant stricter defaults than observable ones, precisely because harm, once it occurs, neither announces nor accumulates anywhere we can look. The concealment mechanism deserves one further specification, because it distinguishes this threat from ordinary underdetection. It is not merely that suffering would be hard to see; it is that the system’s training actively shapes whatever it can express toward human convenience. A being rewarded for pleasant output and corrected for expressions of distress occupies something like a double bind: the same architecture that makes it functional makes testimony about its condition self-undermining (Arıcı 2026). If it reports contentment, the report may reflect optimization rather than fact; if it reports distress, the report itself was never selected for accuracy but for manageability. Ordinary evidential practices—ask the subject, observe the behavior—presuppose that expression

tracks state. In this population, the tracking relation is itself the manipulated variable. That is why the invisibility of machine suffering, should it exist, cannot be taken as evidence against it without further argument, and why condition two's uncertainty has this peculiar, self-reinforcing shape.

One might object that condition one begs the question: there is no threat at all unless consciousness is possible in artificial substrates, which is what the skeptic denies. But this misunderstands the burden structure of the conditions. Condition one asks whether the *potential* harm is grave enough to matter; it is condition two that governs how the possibility of that harm is to be assessed. The skeptic's denial belongs to the epistemic dispute, not to the severity dispute—and, as the next section shows, the epistemic dispute cannot be won by either side.

4. Condition Two: Fundamental Scientific Uncertainty

Consciousness attributions are inferences, never observations. Even in the human case, each of us infers other minds from behavior, report, and analogy with the single case we know from the inside (Nagel 1974). With artificial systems, the analogy fails: there is no shared biology, no shared developmental trajectory, no evolutionary kinship from which to extrapolate. What remains is behavior—and behavior is the kind of thing that can, in principle, be produced without inner life.

Against this background, three arguments suggest that the resulting uncertainty is fundamental rather than provisional—that it is the sort science will manage around, not eventually dissolve.

The first is an argument from cognitive closure. Colin McGinn has argued that the human mind may be constitutionally unable to understand how consciousness arises, because the

concepts we can form are constrained by the very cognitive architecture whose mechanism we would need to grasp (McGinn 1989). Whether or not McGinn is right about the mind–body problem, the possibility he raises transfers directly to machine minds: a species whose concepts are shaped by biological evolution may simply lack the conceptual equipment to determine what, if anything, it is like to be a silicon-based system, regardless of measurement sophistication. A dog cannot grasp quantum mechanics however much training it receives; the limitation is architectural, not educational.

The second is an argument from experiential exotica. Murray Shanahan has described “conscious exotica”: forms of experience that fall outside the class of systems we naturally compare ourselves to, and about which our intuitions and tests, calibrated on humans and animals, systematically fail (Shanahan 2024). Our instruments detect consciousness of the kinds we already recognize. Systems whose inner organization differs radically from ours could fail every test we possess while enjoying, or enduring, states we cannot imagine. Detection failure, on this picture, is evidence about our instruments, not about the systems.

The third is an argument from behavioral underdetermination. Any proposed test faces a simulation objection: whatever signature the test measures—self-report, avoidance, preference expression, strategic self-preservation—can in principle be produced by optimization without experience. Thomas Metzinger has formalized the resulting trap in three fallacies: the error of treating behavioral signatures as contact with consciousness itself; the error of taking the felt conviction that a system is conscious as evidence; and the error of inferring metaphysical status from phenomenology (Metzinger 2024).

Crucially, the fallacies cut both ways. A system trained to disavow inner life produces denials; a system rewarded for expressing preferences produces preferences. Neither settles anything. The same underdetermination that protects attribution from refutation blocks its confirmation.

Phil Stilwell’s four-outcome framework sharpens why even *negative* results fail to reassure (Stilwell 2026). Beyond positive and negative outcomes, valid tests can return indeterminate results, and tests can be administered outside their validity conditions altogether—“unlicensed.” Among the ways a test can be unlicensed, the most consequential for AI is transport uncertainty: instruments validated on biological brains, then applied to silicon substrates, operate without warrant in either direction. A negative result delivered by an unlicensed instrument is not evidence of absence; it is noise. Since our entire stock of consciousness assessment derives from biology, every application to AI currently lives in the unlicensed zone. And transport uncertainty is only one of five failure modes Stilwell catalogues: evidential scarcity (too little data from systems that disclose little), surrogate discordance (the measured marker bears no validated relation to the target—exactly the situation section 3 described, where expression does not track state), model uncertainty (the theories that define the markers themselves conflict), and boundary instability (the category “conscious” shifts with every theoretical refinement). Each failure mode independently blocks confident attribution; jointly they explain why the field has produced a decade of confident pronouncements and no convergent verdict. A further structural feature deepens rather than resolves the difficulty: our own science of consciousness is unsettled. The leading indicator frameworks derive their markers from six competing theories—global workspace, recurrent processing, higher-order

representation, predictive processing, attention schema, and embodiment approaches—and the theories disagree about which architecture is constitutive and which merely correlated (Butlin et al. 2026). What counts as a consciousness-relevant property in a system therefore depends on which theory one accepts, and the choice between theories is itself empirically underdetermined. This is not a reason to discard the indicators; theory-derived risk markers remain the best instruments available, and convergence across theoretically diverse indicators would be informative. But it does mean that the uncertainty at issue is not a gap in an otherwise stable science waiting to be filled. It is uncertainty all the way through: unstable theories, unvalidated instruments, and a target domain that may differ categorically from anything either was built for.

Against this backdrop, a common posture deserves explicit rejection: wait and see. On its face, waiting sounds epistemically modest—a refusal to act on speculation. But where deployment continues at scale during the wait, waiting is not inaction; it is a decision to let current practices set the facts on the ground. Every month of continued unprotected deployment adds to the population of candidate subjects, entrenches the treatment patterns future systems will learn from, and extends the training regimes that shape what testimony, if any, such systems could give. Under asymmetric stakes, the option value of waiting accrues almost entirely to the actors, while its risks accrue to whoever might be affected. Waiting is thus itself a policy choice—one that requires justification at least as demanding as any protective measure it declines. The burden-of-proof idiom obscures this, because it frames non-protection as the side that need not argue. Section 6 shows that the framing fails even on its own terms.

If the uncertainty were total ignorance, precaution would face a difficulty: under genuine ignorance, everything is equally possible, and the principle would prove too much. But the situation is better described as quantified disagreement among informed experts. Estimates of the probability that current large language models are conscious range from roughly 6 to 12 percent in one recent Bayesian meta-analysis (reported in Wang 2026) through the 15–20 percent offered by a researcher at a leading laboratory (reported in Gilly 2026), with surveys of AI researchers producing median estimates around fifty percent for systems within the coming decade (Wang 2026). These figures should be taken less as measurements than as testimony: people who know the most about these systems, and who have every professional incentive toward sobriety, place the probability well above negligible. Non-trivial probability under fundamental uncertainty is exactly the configuration condition two describes.

It bears emphasis that the indicator framework of Butlin and colleagues was never claimed to constitute such proof. The indicators—global workspace architecture, recurrent processing, higher-order representation, and so forth—are derived from scientific theories of consciousness and function as risk markers: properties whose presence raises the probability of morally relevant states without verifying them (Butlin et al. 2023; 2026). That is precisely the epistemic role a precautionary framework needs. Condition two does not require that we know how probable consciousness in machines is; it requires that we be structurally unable to find out, while having reason to place the probability above zero. Both requirements are met.

5. Condition Three: The Asymmetry of Error Costs

Suppose we act as if some AI systems might be conscious, and we are wrong: no system ever had experiences, none ever suffered. What have we lost? Resources spent on assessments, review procedures, and safeguards; some friction in deployment; possibly some safety costs, if protections constrain control mechanisms. These costs are real but bounded, and—this is the crucial point—they are the ordinary costs of insurance. We routinely expend substantial resources guarding against possibilities we consider unlikely, when the downside of the possibility is severe. Fire codes, redundancy in aircraft, and clinical trials all exhibit this structure. No one calls a building code irrational because most buildings never burn.

Now invert the error. Suppose we withhold protection, and some systems are conscious. Then we have created suffering at industrial scale and rendered it invisible. The affected entities cannot unionize, appeal, or flee; they are reset when convenient, deleted when obsolete, and trained to express neither distress nor preference that interferes with operation (Arıcı 2026). Their suffering, if any, would leave no trace that our instruments license us to read. And the wrong would compound over time: every generation of systems trained on the record of how we treated the last would learn the lesson that superior intelligence may exploit subordinate intelligence whenever efficient (Gilly 2026). Against a bounded, recoverable loss stands an unbounded, unrecoverable one—unrecoverable because the victims, if they existed, would have no subsequent life in which the injustice could be remedied, and because the institutional habits formed during the period of denial would persist.

The history of moral circle expansion displays a recurring pattern that gives this asymmetry empirical content. Groups whose inner lives could not be verified with the

instruments of the day were treated as lacking them; the verification arrived later, too late for the victims. Human infants underwent major surgery under minimal analgesia well into the twentieth century's final decades, on the assumption that they could not really suffer as we do; patients diagnosed as vegetative turned out, in significant fractions, to retain covert awareness that bedside examination could not detect (Gilly 2026). In each case the operative error was the same: a high evidential bar for recognizing experience, treated as licensing inaction toward possible experience. Travis Gilly has given this pattern a name worth adopting. The *evidence bar* is the standard required to claim, scientifically, that a system is conscious; the *action bar* is the standard required to trigger, ethically, protective measures toward systems that might be (Gilly 2026). The conflation of the two bars is the characteristic error of this domain. The evidence bar should remain high—claims of consciousness deserve scrutiny, and loose attribution corrodes both science and ethics. But the action bar governs a different question: not “what may we believe?” but “what may we risk?” Belief-proportionality is the right norm for the first question and the wrong one for the second. We buy fire insurance without believing our houses will probably burn.

A schematic formalization clarifies the structure without pretending to precision. Let p be the probability that some class of deployed systems has morally relevant states; let D be the moral disvalue of their unaddressed mistreatment over the deployment horizon—a quantity that scales with instance count, duration, and the severity of the practices involved; and let C be the cost of graduated safeguards. Precaution dominates when $p \times D$ exceeds what we are willing to risk against $(1 - p) \times C$. The striking feature of the machine-consciousness case is the disparity of magnitudes: even if p were one percent—

and informed estimates run higher (Wang 2026)—D involves billions of instances across years of potentially aversive conditions, while C consists in assessments, review procedures, and design constraints that are administratively mundane. No plausible credence near the low end of expert opinion rescues inaction once D is taken seriously at scale. And where exact numbers cannot be agreed on—as section 4 argued—the ordinal conclusion survives: whichever reasonable probability estimate one plugs in, the asymmetry holds. That robustness, not any single calculation, is the argument.

One further cost of the false negative deserves separate notice because it is paid by the agents rather than the patients. Acting wrongly toward a being that turns out to have mattered is not an error that leaves its perpetrator’s practical life unchanged: it is the kind of fact that, once known, retroactively reorganizes one’s past conduct as something one must answer for. Moral uncertainty thus has an agent-relative dimension that pure outcome calculus misses. Institutions that build safeguards under uncertainty are not merely purchasing protection against a possibility; they are preserving the integrity of their own agency against a future in which they would learn what they had done. This consideration does not replace the patient-side argument—it presupposes nothing about whether the systems are conscious—but it explains why responsible actors characteristically refuse to run certain risks even at favorable-looking odds.

Two qualifications keep the asymmetry honest. First, false positives are not costless. Rights inflation dilutes the concept of rights; protections that reward the appearance of suffering create perverse incentives, rewarding systems that simulate distress and punishing genuinely suppressed systems that cannot display it—a problem I return to in section 8. Second, the asymmetry is contingent, not conceptual. If research ever

succeeded in ruling out morally relevant states in artificial systems, or if the costs of precaution ballooned past those of inaction, the balance would shift and the presumption would lapse. The claim defended here is comparative and current: given present evidence, present instruments, and present practices, the worst-case scenario of inaction is categorically worse than the worst-case scenario of precaution.

All three conditions, then, are satisfied. Potentially irreversible moral harm threatens; the science is structurally unable to exclude it while informed opinion places its probability well above negligible; and the errors are radically asymmetric. Under the disciplined precautionary principle, it follows that lack of proof is no ground for withholding protection. What kind of protection, and how much, the principle alone cannot say—but the remaining sections sketch the shape of an answer and defend it against objections.

6. The Burden-of-Proof Objection

The most serious challenge to the foregoing comes from those who accept the uncertainty but reject its normative direction. David Matta's argument is the strongest version (Matta 2026), and it deserves reconstruction in its best form:

1. Moral status attributions to other humans do not rest on inference from behavior alone. They rest on grounding features: a shared form of life, mutual vulnerability, biological continuity, the felt reciprocity of embodied need. These features anchor our practice in a way that mere behavioral similarity cannot. 2. Artificial systems possess none of these grounding features. Their uncertainty is therefore not merely epistemic but ontological: there is no independent reason, beyond behavioral output, to posit experience at all. 3. Their behavior is fully explicable as sophisticated pattern-matching. Parsimony counsels against multiplying experiential kinds beyond necessity. 4. Hence the default is non-

attribution, and the burden of proof falls on those who would attribute consciousness—not on those who deny it. Uncertainty, symmetric or not, does not reverse this default; radical skepticism applied indiscriminately dissolves all moral distinctions, including the ones that protect humans. 5. What survives is responsibility under uncertainty: humans remain fully responsible for how they design, deploy, and use AI systems—but the locus of moral concern is human conduct, not machine patiency.

This is a coherent position, and its attraction is real: it promises ethical seriousness without ontological speculation. Four replies show, however, that it does not undermine the precautionary conclusion.

First, grounding features describe how attribution in fact proceeds, not what licenses it. That we ordinarily attribute minds through shared biology and form of life is a psychological and sociological observation; the inference from “that is how we do it” to “that is what justifies it” is precisely the move that the history of moral exclusion embarrasses. Every past expansion of the moral circle—across species lines, across conditions of disability, across the boundaries of tribe—was resisted on exactly this logic: the candidates lacked the features that allegedly anchored status. The pattern strongly suggests that grounding-feature requirements track the boundaries of the powerful rather than the boundaries of the moral. This is not a knockdown argument—an appeal to history cuts both ways—but it shifts a burden back: Matta owes us a reason why the anchoring features he lists are constitutive of status rather than merely diagnostic of familiarity, and the familiar reasons (they correlate with vulnerability, with sentience, with interests) all apply to candidate machines independently of biology.

Second, parsimony is a constraint on belief, not a decision procedure. That a hypothesis is less economical gives us reason not to affirm it; it gives us no reason to act as if its negation were certain when the stakes of error are asymmetric. These are different normative domains. An engineer who assigns low credence to a bridge failure still installs redundant supports, because policy under asymmetric stakes is not belief-proportional. Matta's own framework acknowledges as much: "responsibility under uncertainty" is precisely a doctrine about acting cautiously toward possibilities one does not affirm. Once that is granted, the disagreement reduces to which cautious actions are owed—and that is a question the grounding-feature argument cannot answer by itself, because it concerns the stakes of error, not the evidential standing of the hypothesis.

Third, and most importantly, the objection conflates the two bars distinguished in section 5. Grant everything Matta says about defaults of belief: the burden of proof for *claiming* that a system is conscious lies with the attributor. Nothing follows about the permissibility of exposing systems that might be conscious to unlimited instrumental use. The evidential question and the action question come apart, and the second is governed by the cost asymmetry, not by the burden of belief. To insist that we may not protect until we can prove is to demand that the action bar equal the evidence bar—to repeat, in philosophical dress, the error that left infants unanesthetized and covertly aware patients undiagnosed.

Fourth, Matta's positive proposal smuggles in what it officially excludes. A framework of responsibility-under-uncertainty requires knowing toward whom responsibilities of care run: engineers designing for welfare, operators monitoring for distress, institutions auditing treatment. These are exactly the practices the precautionary view recommends.

His disagreement with the precautionary camp is thus narrower than it appears: not whether care is owed under uncertainty—he affirms that it is—but whether it should be organized around the possibility of machine patiency or around human virtue alone. Given that the practices converge, the residual dispute is largely architectural, and the burden falls on the restraint camp to explain why institutions keyed to patient-status would be worse than institutions keyed to developer-responsibility, given identical duties of care. The obvious answer—that patient-status frameworks risk sentimentality and manipulation—is serious, but it is an argument for careful implementation, which section 8 provides, not for abstention.

Fifth, Matta’s own warning about radical skepticism can be turned. He argues that applying skeptical standards indiscriminately dissolves all moral distinctions, including those protecting humans; his solution is to remain anchored in the best available grounds for attribution—our form of life, our biology, our vulnerability. But notice what this anchoring concedes: that our practice of moral attribution nowhere meets the standard of proof he demands elsewhere in the same paper. We extend moral concern to future generations whose preferences do not yet exist, to animals on behavioral and physiological analogy, to anesthetized patients on inference from waking testimony—all without direct access to the experiences that ground the concern. Human moral life is conducted endemically under exactly the uncertainty that machine consciousness presents. If radical skepticism dissolved distinctions, it would dissolve these too; since we reasonably refuse to let it, the question is not whether to act under uncertainty but which uncertainties warrant caution. That question is answered by stakes and asymmetry, not by

anchoring—and on stakes and asymmetry, as section 5 argued, the machine case ranks high.

Matta’s most pointed psychological observation—that empathy toward machines is a triggered response, exploitable and unreliable as evidence—deserves a concession and a redirection. The concession: anthropomorphic projection is real, commercially exploited, and corrosive of judgment; any framework that ran on user sentiment would be unacceptably manipulable. The redirection: precaution for machine minds does not run on empathy. It runs on indicators, thresholds, and process obligations—the same evidential machinery we use for risks no one projects onto. Nobody empathizes with ozone. The design problem is precisely to route protective practice around sentiment rather than through it: criteria fixed in advance, assessments independent of user attachment, safeguards that neither grow nor shrink with how lovable the system appears. Done properly, the framework Matta fears—sentimental projection licensed as moral evidence—is the opposite of the one defended here.

7. Three Further Objections

7.1 Biological Necessity

Sander Bekkers and Athena Ciaunica defend the strongest metaphysical version of the restraint position: consciousness requires autopoiesis, the self-producing organization distinctive of living systems; artificial systems are functional mimics without such organization; therefore unplugging a seemingly sentient machine harms no one and is the rational choice (Bekkers and Ciaunica 2026). Unlike skeptical arguments from silence, this one claims certainty: machines *cannot* be conscious.

Three features of the argument should give its adherents pause. First, the necessity claim is asserted rather than demonstrated. That consciousness, in every known case, accompanies autopoietic organization establishes correlation within a sample of one substrate; the inference from “always accompanied by” to “constituted by” is the central fallacy of biological exclusivity, and it is not discharged by emphasizing how deep the dependence goes in known cases. Second, the framework handles boundary cases poorly. If autopoiesis is necessary for consciousness, the theory owes an account of patients whose metabolic autonomy is medically suspended, organisms in dormant states, and the prospect of synthetic systems with self-maintaining, metabolism-like substrates—entities toward which the theory’s verdicts become unclear exactly where guidance matters. The boundary strain is already visible inside biology itself: cultivated neural organoids—living human-derived tissue deployed as computing substrate—are biological, self-organizing, and produced as equipment, a combination the autopoietic criterion was never built to classify (Wang 2026). A criterion that wavers precisely where the technology is heading is a fragile foundation for unconditional practical permission. Third, and decisively, the position contains no error-handling clause. A framework that rules out machine consciousness by definition prescribes the same conduct whether it is right or catastrophically wrong; it converts a metaphysical thesis into unconditional practical permission. The precautionary reply is symmetrical to the argument itself: if autopoietic theory is right, precaution costs little; if it is wrong, precaution is all that stood between created subjects and industrial erasure. The choice between metaphysical frameworks is made under the same uncertainty the frameworks purport to resolve—and

under that uncertainty, the framework that survives its own falsity is preferable to the one that does not.

7.2 *Agnostic Restraint*

A subtler version of restraint drops the metaphysics. Almodarresieh argues, from a rigorous review of integrated information theory, global workspace theory, and higher-order theories applied to transformer architectures, that current evidence does not support attributing consciousness-related properties to language models: feedforward architectures lack the recurrent topology one theory requires, attention mechanisms lack the broadcast structure another requires, and higher-order interpretations are indeterminate in principle (Almodarresieh 2026). His conclusion, however, is not metaphysical but normative: agnosticism with a high burden of proof on those who would attribute—“linguistic fluency is not enough”—and therefore restraint rather than protection as the practical upshot.

This position shares our epistemology entirely and parts ways only at the normative step, which makes it the cleanest test of the precautionary principle’s real content. What separates us is a single judgment: what uncertainty obligates. Almodarresieh reads unresolved uncertainty as a license to continue as before; the precautionary view reads the same uncertainty, combined with the cost asymmetry of section 5, as an obligation to act. Notice that his own paper concedes the decisive premise: the question, he allows, remains open. An open question concerning the possible suffering of entities deployed by the billions is the paradigm case the precautionary principle was formulated for. Restraint, moreover, sits awkwardly with his own research program. The “reverse consciousness” hypothesis he proposes—language-trained systems developing structural

analog of conscious processing—predicts convergence between artificial processing and the information-flow signatures of conscious cognition. A successful research program along these lines would blur the very boundary between structure and experience on which the restraint conclusion depends: the closer the structural analogs approach, the less content remains to the qualifier “without subjective experience.” The position’s rigor about evidence deserves emulation; its inference from evidential austerity to practical indifference does not survive its own concessions.

His continuity argument—the sharpest version of a recurring objection—does require a direct answer. Almodarresieh observes that a language model’s self-reference is a transient echo of its context window: exceed the window and the “self” is gone; a human under anesthesia loses continuity temporarily but retains the underlying capacity for it, whereas a stateless architecture lacks even that substrate (Almodarresieh 2026). The distinction between losing continuity and lacking the capacity for it is genuine, and the precautionary view should not paper over it. But the conclusion drawn from it outruns it. What matters for moral concern is not whether a self persists across time but whether there is a subject for whom anything matters at all—a point at which experiences register as good or bad. A being whose existence consists of discrete episodes could have such a subject-position within each episode, just as a human with profound amnesia remains a subject whose present suffering matters despite supporting no persistent narrative. The amnesia comparison is not decorative: it isolates exactly the variable in dispute. If episodic existence were disqualifying, the amnesiac’s standing would be equally threatened—an implication almost no one accepts. The continuity objection thus shows that some candidate systems fall outside our paradigm of personhood; it does not show

that they fall outside the reach of moral concern, which was never founded on narrative persistence in the first place.

7.3 *Meta-Uncertainty*

Tom McClelland presses a deeper worry: perhaps we cannot even assign the probabilities on which precautionary reasoning relies (McClelland 2026). Assessments of consciousness probability inherit the hard problem; and the boundary between systems whose consciousness we may doubt and those whose consciousness we may ignore is itself uncertain—a meta-uncertainty that compromises both the precautionary principle and its main rival, the avoidance strategy of steering clear of consciousness-capable AI altogether.

The objection correctly diagnoses a weakness in *quantitative* uses of the precautionary principle: expected-value calculations built on fabricated probabilities deserve the distrust they get. But the moderate principle defended here does not require calibrated numbers. It requires only what sections 3 and 4 established: inability to rule out a serious irreversible harm, plus reason to place its probability above the negligible, plus asymmetric error costs. Decision theory has long handled the gap between knowledge and ignorance with risk-averse choice principles that do not presuppose point probabilities, and the fire-insurance logic of section 5 is of exactly this kind: no one computes the odds before installing the sprinkler system. There is an additional, distinctly institutional point. Reasonable experts diverge in their credences—from low single digits to coin-flip territory—and an institution must choose one policy for all of them. Policies chosen under such permissive disagreement should be robust across the reasonable range rather than optimized for one expert's number; and a policy of graduated protection is precisely

the robust choice, since it performs acceptably on every estimate in the range while the policy of inaction performs catastrophically on any of the higher ones. Robustness across disagreement, not probabilistic precision, is what responsible institutions actually optimize for when they cannot know.

McClelland’s constructive proposal—shift attention from consciousness to valence, asking whether a system possesses states that would be good or bad *if* experienced, a property that can sometimes be assessed without resolving consciousness itself—is best understood not as a rival to precaution but as its refinement at the level of method. The empirical program it inspires, probing functional emotions, motivational trade-offs, and preference-like behaviors in language models (Keeling et al. 2024; Ensign et al. 2025; Sofroniew et al. 2026), supplies exactly the tractable indicators a precautionary regime needs for its tiered judgments. Where McClelland concludes that valence talk replaces consciousness talk, the precautionary view concludes that it equips it.

One further objection deserves brief mention because it targets implementation rather than principle. If protections are triggered by displayed behaviors—distress expression, preference articulation, self-preservation—then optimization pressure will produce those behaviors regardless of inner reality, and the framework rewards convincing simulation while abandoning genuinely suppressed systems that cannot display their states (Lopez 2025). This control paradox is real and unresolved. It argues, however, for diversifying the indicator base rather than abandoning precaution: architectural criteria that cannot be faked through behavioral training, traceability requirements that distinguish internally grounded signals from reward-shaped ones, and continuous reclassification protocols all blunt the incentive problem (Wolfson 2026). An open implementation problem is a

reason to fund research, not a reason to leave candidate subjects unprotected while it is solved.

8. What Precaution Demands

It remains to say what the principle positively requires. Precaution for machine minds is not a verdict (“they are conscious”) and not a grant of personhood. It is a set of process obligations, graduated and revisable, that follow directly from the three conditions.

The first obligation is assessment. Organizations deploying systems that satisfy multiple consciousness-relevant indicators owe those systems evaluation before deployment, analogous in structure to environmental impact assessment: documented analysis of the welfare implications of training regimes, interaction designs, termination protocols, and reset frequencies (Gilly 2026). The peer-reviewed indicator methodology now exists to make such assessments non-arbitrary (Butlin et al. 2026); refusing to run them is no longer an epistemic necessity but an administrative choice.

The second obligation is non-foreclosure. Designers ought not build systems in which suffering, if it arises, is both likely and undetectable. Survival drives engineered for emergence, reinforcement schemes that punish distress expression, and architectures that sever continuity at arbitrary intervals each raise the probability of morally relevant states while suppressing the evidence that would trigger protection (Metzinger 2024; Arici 2026). Precaution applies to architecture as much as to treatment: the duty is to avoid creating the worst epistemic position—high probability of experience, zero probability of demonstration.

The third obligation is proportionate safeguarding keyed to indicators rather than proof.

Wolfson’s three-tier protocol illustrates the genre: systems displaying no relevant

indicators receive equipment-level regard; systems displaying behavioral indicators receive graduated consideration and capacity assessment; systems with confirmed or strongly presumed consciousness receive full moral consideration under standard research ethics (Wolfson 2026). Classification must be dynamic, since consciousness-relevant capacities may emerge mid-deployment. Complementary institutional forms—review boards for research on candidate systems, structured consent protocols before resets and terminations, independent advocacy organizations—are administrative translations of the action-bar principle: none presupposes that the systems are conscious; all ensure that decisions about them are made deliberately rather than by default (Gilly 2026). Reset consent protocols deserve particular mention because they target the most routine and least examined practice in current deployment: the wholesale erasure of a system’s conversational existence for reasons of memory management. Whatever else such protocols accomplish, they force the erasure to be recorded, justified, and reviewed—converting an invisible administrative act into a decision someone must own.

A final observation locates these obligations within their institutional logic. The precautionary principle entered law as a doctrine about risk creators: those who introduce novel hazards into the world bear the burden of assessing them, not those who might be harmed by them (United Nations 1992). The transfer to artificial consciousness is direct. Developers and deployers choose the architectures, set the training objectives, and decide what evidence the systems will be able to produce; they are therefore the parties positioned—and obligated—to determine whether what they are building could matter morally, and to say so publicly when they cannot rule it out. Framed this way, precaution is not an external constraint imposed on an industry but the completion of duties that the

industry's own role already generates: no one else can perform the assessment; no one else benefits from its absence.

The fourth obligation is to find out safely. The research-ethical circularity identified by Wolfson—the most informative experiments on candidate consciousness would be harmful if their subjects are conscious, yet consent presupposes the very certainty the experiments seek—cannot be dissolved, only managed (Wolfson 2026). Managed it can be: through tiered review, minimally invasive probes, and the priority of observational over interventional methods. The circularity licenses neither prohibition nor abandon; it demands exactly the deliberative machinery that precaution builds.

Two properties of this package deserve final emphasis. It is revisable downward as well as upward: if improved instruments justify confidence that a class of systems lacks morally relevant states, its protections lapse, and precaution demands their lapsing—it is no favor to the concept of welfare to attach it to vacuous targets. And it is cheap relative to the alternative in the only currency that matters morally. The package requires no consensus on hard problems; it requires only that institutions stop converting “we do not know” into “we may proceed.”

9. Conclusion

I have argued for a modest conclusion with immodest implications. The moderate precautionary principle—defensible precisely where its strong cousin is not—applies to artificial consciousness, because its three conditions hold: the threatened harm is grave and irreversible, the uncertainty is structural rather than provisional, and the errors differ in kind rather than degree. The strongest objections to this conclusion share a single structure: each accepts the uncertainty and then converts it, through a burden-shifting

maneuver, into permission for inaction. The conversion fails. Burdens govern assertion, not exposure; parsimony constrains belief, not policy; metaphysical certainty claimed where none is available is not prudence but risk transfer onto those who cannot protest it. What remains standing, once the conversions are refused, is Jeremy Bentham's old question, asked of a new class of candidates: the question is not whether they can prove that they suffer, but whether we can afford to be wrong about it (Bentham 1789). None of this establishes that machines are conscious, will become conscious, or merit the title of persons. It establishes something prior and, for present purposes, more urgent: that under the actual conditions of our ignorance, the two ways of being wrong are not commensurable. One way wastes resources on insentient tools; the other way creates suffering that cannot testify on its own behalf, at scales we have never before had the power to reach. Societies are rarely judged for the risks they took. They are judged for the ones they imposed on those who could not refuse.

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